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SARDAR PATEL UNIVERSITY

S.Y.B.C.A. 4th SEMESTER CBCS Examination April - 2023

US04CBCA52 || Data Structure - II

Date: 06/04/2023 Thursday

Time: 02:00pm to 05:00pm

Maximum Marks: 70

Q.1

Answer the following questions.

[10]

- 1 _____ contains maximum possible number of nodes in all level except possibly the last level.
(a) Full binary tree (b) Skew binary tree (c) Complete binary tree (d) None
- 2 Leaf node is known as _____ node
(a) Terminal (b) Parent (c) Root (d) Non-terminal
- 3 _____ is a set of disjoint trees.
(a) Path (b) Forest (c) Directed tree (d) Height
- 4 A linked list is _____ type of data structure.
(a) Non-Linear (b) Linear (c) Both (a) and (b) (d) None of the Above
- 5 The minimum number of fields with each node of double linked list is _____.
(a) One (b) Two (c) Three (d) Four
- 6 A linked list is considered as an example of _____ type of memory allocation.
(a) Compile time (b) Static (c) Dynamic (d) None of the Above
- 7 _____ is the process to find the particular element from the array.
(a) Sorting (b) Inserting (c) Updating (d) Searching
- 8 Quick sort is also known as _____.
(a) merge sort (b) heap sort (c) bubble sort (d) None of these
- 9 The lowest level of index is _____.
(a) Track index (b) Index area (c) Prime index (d) Master index
- 10 A _____ which defines a mapping from key space to the address space.
(a) Hashing Function (b) Blocking (c) Deblocking (d) None of these

Q.2

Explain following in brief. (Attempt any Ten)

[20]

- 1 Define Following Terms: Tree, Root.
- 2 Define Traversals. List the types of binary tree traversal.
- 3 List out Applications of Tree.
- 4 Explain doubly linked list in brief.
- 5 Explain circular linked list in brief.
- 6 Explain linked list in brief.
- 7 Define searching. List out the names of searching techniques.
- 8 Advantages and Disadvantages of Bubble sort & Merge Sort.
- 9 Define: Internal Sort and External Sort.
- 10 Define Address Space and Key Space
- 11 Explain in brief prime area.
- 12 Define Bucket Capacity.

[P.T.O.]

Q-3 (A) Define Binary tree. Draw a binary tree for following equation. [05]

(1) $(A+B^*C)/(A+B)^*C$

(2) $A+(B^*C)+D^*(E+F)$

(B) Write down an algorithm for preorder traversal of a binary tree [05]

OR

Q-3 (A) Draw the binary tree for following expressions: [05]

Inorder : D B H E A I F J C G

Preorder : A B D E H C F I J G

(B) Explain the insertion of a node in a binary with an example. [05]

Q-4 (A) Write an algorithm to insert node at any position in singly linked list. [05]

(B) Write an algorithm to delete node from end position in singly linked list. [05]

OR

Q-4 (A) Write an algorithm to insert node at front position in singly linked list. [05]

(B) List and explain any two types of linked lists with example. [05]

Q-5 (A) What is sorting? List sorting techniques and explain ANY ONE of them. [05]

(B) Write down the algorithm of Binary Search. [05]

OR

Q-5 (A) Write down the algorithm of bubble sort. [05]

(B) Differentiate between Sorting and Searching. [05]

Q-6 Write a detail note on: Collision resolution techniques [10]

OR

Q-6 List and explain popular hashing function with example. [10]

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